Preparation dependence of proper-time volume in Einstein–scalar shell perturbations

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Abstract

A first perturbation and its quadratic sources need not determine the sign of a second observable when physical homogeneous second initial data remain free. We give a conditional quantitative example for a five-dimensional Einstein–scalar spacetime with a scalar-dependent pure-tension shell. A compact zero-moment metric mean changes the second preparation while preserving the first quadrupole packet, canonical constraints, initial shell collar, and second transverse-traceless initial pair. The difference solves the complete homogeneous five-mean problem. At one fixed anchored proper time, its proper-volume and expansion slopes are strictly negative. The fixed finite preparation parameter $\lambda = -10^{21}$ therefore changes both second coefficients from negative to positive. Whole-cell rational enclosures give $v_2 \in [1.0343, 1.8442] \times 10^{17}$ and $\Theta_2 \in [8.3831, 18.496] \times 10^{21}$; a separate elementary bound also proves both signs. The result assumes the stated constraint-inverse, development, regularity, and linear-uniqueness premises. It supplies no effective nonlinear parameter radius, contracting branch, or physical rule selecting a preparation. The affine structure is established perturbation machinery; the contribution is this model-specific signed example.

1 Model, fixed data, and conditional result

Bulk initial information can survive in brane observables [1, 2, 3]. Ordinary second-variation theory also separates quadratic forcing from homogeneous second data. The question here is more specific: can two compact, constraint-compatible preparations with the same first packet and initial shell collar give opposite signs for both proper-volume and expansion second coefficients at the same event? Non-gauge data alone do not answer it; their observable response must be shown to be nonzero. We construct one direction and bound that response.

The bulk and one-copy outward shell equations are

$$\mathrm{Ric}_{AB} - \phi_A\phi_B - \tfrac{2}{3}\mathcal{V}(\phi)g_{AB} = 0, \qquad \Box_g\phi - \mathcal{V}'(\phi) = 0, \tag{1}$$

$$K^{\mathrm{out}}_{ab} = \tfrac{1}{6}\sigma(\phi)\gamma_{ab}, \qquad n_{\mathrm{out}}\phi = -\tfrac{1}{2}\sigma'(\phi). \tag{2}$$

Two identical interior copies use their respective outward normals. Fix

$$W(\phi) = 1 - \phi + \phi^3/3, \qquad \mathcal{V} = \tfrac{1}{2}(W')^2 - \tfrac{2}{3}W^2, \qquad \sigma = 2W + \delta(1 + c_\sigma\phi),$$
$$\delta = 10^{-3}, \qquad c_\sigma = 96714321425692/161839258930731. \tag{3}$$

The selected near-$\phi = 1$ regular background is

$$g_0 = \mathrm{d}y^2 + \rho(y)^2[-\mathrm{d}t^2 + \cosh^2 t\,\kappa], \qquad a = \rho_y/\rho, \quad p = \phi_{0,y}, \tag{4}$$

where κ is the unit round metric on S^3. Its exact selected shell radius is ρ_b; it is not a rounded value chosen from a bounding box. Put $\ell = 10^{-6}$, $x = t/\ell$, $r = z/\ell$, and $\mathrm{d}y = -\rho\,\mathrm{d}z$. The packet support is $1 \le r \le 2$. The fixed target is $T = 5/4$, or $\tau_* = \rho_b \ell T$ in background proper time. All quantities below are in these model units.

We use second parameter derivatives throughout:

$$x_\varepsilon = x_0 + \varepsilon v_{\mathrm{seed}} + \tfrac{1}{2}\varepsilon^2 x_2 + o(\varepsilon^2), \qquad g_\varepsilon = g_0 + \varepsilon q + \tfrac{1}{2}\varepsilon^2 k + o(\varepsilon^2). \tag{5}$$

Anchor the shell's intrinsic future normal-geodesic foliation at the common initial section, with induced spatial metric $h_\varepsilon(\tau)$, and set

$$V_\varepsilon = \int_{S^3} \mathrm{d}\mu_{h_\varepsilon}, \quad \Theta_\varepsilon = \partial_\tau \log V_\varepsilon, \quad M_\varepsilon = V_\varepsilon^{-1} \int_{S^3} \phi_\varepsilon \,\mathrm{d}\mu_{h_\varepsilon}, \quad v_2 = V_2/V_0. \tag{6}$$

The clock is intrinsic to the shell and transported under gauge changes. The background has $A(\tau) = \rho_b \cosh(\tau/\rho_b)$, $H = A_\tau/A$, and $V_0 = 2\pi^2 A^3$.

The analytic premises are explicit. We assume the selected regular background and its authenticated annulus bounds; the equivariant weighted constraint isomorphism and nonlinear implicit-function hypotheses; the common fixed-shell development and subsidiary/boundary uniqueness; and identification of its first derivative with the retained first-field solution. For each fixed finite preparation, assume C^2 parameter dependence in the retained lower C^4 spacetime topology, with the traces and derivatives required in (6). A common intrinsic normal tube reaches the target after a family-dependent qualitative shrinkage in ε. The full linear mean mixed problem is unique and obeys the coefficient bounds in Section 4. These analytic premises are retained from the technical development [9]; they are not proved by finite interval arithmetic. No effective admissible ε radius is assumed or obtained.

Theorem 1 (Conditional preparation dependence)**.** *Under these premises, there are two exact completed families with the same background, first packet, canonical quadratic constraint source, initial shell collar, and second transverse-traceless initial pair, for which the second coefficients at the same proper clock satisfy*

Preparation	$v_2(\tau_*)$	$\Theta_2(\tau_*)$
Base	$[-5.4520, -2.6521]\,10^{16}$	$[-9.0208, -4.4458]\,10^{21}$
$\lambda = -10^{21}$	$[1.0343, 1.8442]\,10^{17}$	$[8.3831, 18.496]\,10^{21}$

These are outward decimal displays of exact rational enclosures. Consequently the retained first-packet specification does not determine either sign. This single direction proves simultaneous reversal, without asserting independent control of the two coefficients.

2 A compact constraint-compatible preparation direction

Here are the fixed first data and the base preparation, so that the comparison identifies what is held constant. In a background orthonormal spatial frame, with background spatial density ρ_0, the first canonical data for the action normalization $R/2 - |\mathrm{d}\phi|^2/2 - \mathcal{V}$ are

$$\gamma_1 = \mathrm{diag}(-4u, 2u, 2u, 2u)Y, \quad \pi_1/\rho_0 = \mathrm{diag}(-3s_1/2, 0, 0, 0)Y, \quad \phi_1 = vY, \quad \varpi_1/\rho_0 = jY. \tag{7}$$

The angular factor is $Y = n_1^2 - n_2^2$, with $\langle Y\rangle = 0$, $\langle Y^2\rangle = 1/6$, and $\Delta_\kappa Y = -8Y$; brackets use probability measure. Define

$$\begin{aligned} Q(r) &= \exp\!\left(4 - \frac{1}{(r-1)(2-r)}\right) \quad (1<r<2), \qquad Q = 0 \text{ otherwise}, \\ B(z) &= Q(z/\ell), \quad \alpha = (\rho/\rho_b)^{3/2} p/p_b, \quad \varkappa = \alpha_z/\alpha = \rho(5a/2 - \mathcal{V}'/p), \\ u &= \alpha B/\rho^2, \quad u_t = e^{-z}u, \quad s_1 = u_t/\rho, \\ v &= 3\alpha(B_z + \varkappa B)/(\rho^3 p), \quad v_t = 3\alpha e^{-z}[B_z + (\varkappa - 1)B]/(\rho^3 p), \quad j = v_t/\rho. \end{aligned} \tag{8}$$

All endpoint extensions are smooth and flat. In particular $pv = -3(u_y + 2au)$ and $pj = -3((s_1)_y + 3as_1)$. The mean Hamiltonian and radial momentum Hessians are

$$\begin{aligned} Q_H = &-2u_y^2 - 24auu_y - 2uu_{yy} - (16a_y + 32a^2 + 8p^2/3)u^2 - 4u^2/\rho^2 - s_1^2 \\ &- (j^2 + v_y^2 + \mathcal{V}''v^2)/6 - (4/3)puv_y - (4/3)v^2/\rho^2, \\ Q_M = &-2s_1u_y + j(v_y + 4pu)/3. \end{aligned} \tag{9}$$

They are Hessians in convention (5). Let $y_L = y(2\ell) < y_U = y(\ell)$ and

$$\begin{aligned} I = \int_{y_L}^{y_U} \rho^4 Q_H \,\mathrm{d}y, \quad J = \int_{y_L}^{y_U} \rho^2 B \,\mathrm{d}y > 0, \quad f = \frac{2I}{9J}B, \\ C[f] = -\tfrac{3}{2}(f_{yy} + 3af_y) - \mathcal{V}f, \qquad \xi = -\frac{\int_{y_L}^{y} \rho^4(Q_H + C[f])\,\mathrm{d}s}{\rho^4 p^2}. \end{aligned} \tag{10}$$

The base second mean is the canonical datum

$$B_{\mathrm{base}} = \Big(f\gamma_0 + \mathcal{L}_{\xi\partial_y}\gamma_0, \rho_0 \,\mathrm{diag}(0, -Q_M/(6a), -Q_M/(6a), -Q_M/(6a)), 0, 0\Big). \tag{11}$$

It solves the second mean constraint equation $PB_{\mathrm{base}} = -P_0\mathcal{Q}$, where $\mathcal{Q}$ is the complete first constraint Hessian and P_0 is Haar averaging. The base numerical response in Theorem 1 retains the actual first evolution and shell forcing, not merely (9).

Now let h be compact and flat in the same packet interval, with

$$\int_{y_L}^{y_U} \rho^2 h \,\mathrm{d}y = 0, \qquad h = \rho^{-2}\chi_y, \qquad \chi(y) = \int_{y_L}^{y} \rho^2 h \,\mathrm{d}s. \tag{12}$$

For every fixed finite real λ, set

$$f_\lambda = f + \lambda h, \quad \xi_\lambda = \xi + \lambda\zeta, \qquad \zeta = \frac{3(h_y - ah)}{2p^2} + \frac{9\chi}{2\rho^4 p^2}. \tag{13}$$

The canonical momentum mean and both scalar canonical sectors in (11) are unchanged. Thus $B_\lambda - B_{\mathrm{base}} = \lambda(H_h, 0, 0, 0)$, where $H_h = h\gamma_0 + \mathcal{L}_{\zeta\partial_y}\gamma_0$. The background identity $a_y + 4a^2 = 3/\rho^2 - 2\mathcal{V}/3$ gives the exact flux formula

$$\rho^4 C[h] = -\tfrac{3}{2}[\rho^4(h_y - ah)]_y - \tfrac{9}{2}\rho^2 h, \quad P_g H_h = C[h] + \rho^{-4}(\rho^4 p^2 \zeta)_y = 0. \tag{14}$$

The moment condition makes its primitive zero on both exterior sides. Flatness and $p \neq 0$ give compact smooth extension. This changes no part of the quadratic source: $PB_\lambda = -P_0\mathcal{Q}$ exactly.

For fixed retained weights, insert $\varepsilon^2 B_\lambda/2$ into the same weighted implicit-function trial $x_\varepsilon = x_0 + \varepsilon v_{\mathrm{seed}} + \varepsilon^2 B_\lambda/2 + \mathcal{A}U_\lambda(\varepsilon)$. The correction derivative is the same isomorphism $L = \omega^{-2}P\mathcal{A}$, and

$$LU_\lambda''(0) = -\omega^{-2}(\mathcal{Q} + PB_\lambda). \tag{15}$$

It is independent of λ. Retained equivariance and injectivity give $P_0 U_\lambda''(0) = 0$, hence exactly the declared second mean. Compact support preserves the initial shell collar and second TT pair. The common first tangent and retained linear uniqueness preserve the first field and all its quadratic sources. For every fixed finite λ, the existence and support arguments apply after qualitative shrinkage in ε. No radius uniform as $|\lambda|$ grows, no choice $\lambda = \lambda(\varepsilon)$ tending to infinity, and no affine full finite-ε family are claimed.

The freedom is physical in the anchored pure second gauge quotient with the common first representative fixed. On the time-symmetric initial section $K_0 = \Pi_0 = 0$, so a normal generator changes neither initial spatial metric nor scalar. Averaging a proposed tangential generator leaves a radial generator; there is no invariant angular one-form. Its scalar variation is pX^y. The raw scalar difference is zero, so $X^y = 0$. For completeness, $H_h = 0$ would imply $h + 2\zeta_y = h + 2a\zeta = 0$, hence $\zeta_y = a\zeta$; compact support then gives $\zeta = h = 0$. A nonzero h is therefore not such a diffeomorphism. This does not imply that every observable detects it.

3 Conditioning, proper observables, and exact affine response

For each preparation a pure second harmonic vector has initial data $(Z^y, Z^t, Z^y_t, Z^t_t) = (-\xi_\lambda, 0, 0, a\xi_\lambda)$ and shell rows $Z^y = 0$, $\partial_y Z^t = 0$. It solves $(\Box_0 + \mathrm{Ric}_0)Z = 0$ and transports the shell anchor and clock. The transformation $\bar{k} = k + \mathcal{L}_Z g_0$, $\bar{w} = w + Z\phi_0$ removes the spatial Lie term and keeps its essential scalar pullback. It is a change of representative of each family, not the physical preparation freedom just constructed. The second jet suffices; no nonlinear harmonic transformation at finite ε is asserted. The factorial convention and transport follow the standard second-order gauge framework [4, 5].

In the background orthonormal frame, let (K, L, N, J, w) denote the $yy, y0, 00$, angular-isotropic, and scalar mean slots. Subtracting the conditioned adapters gives the initial data for the unit direction:

$$\begin{aligned} \mathbf{W}(0,r) &= (h,0,0,h,s), \qquad s = -p\zeta, \\ \mathbf{W}_x(0,r) &= (0, \ell\rho(-h_y + ah), 0, 0, 0), \qquad \mathbf{W}_r(0,r) = (h_r, 0, 0, h_r, s_r). \end{aligned} \tag{16}$$

The mixed metric velocity is retained. The scalar coordinate velocity is zero; the first-only lapse-square term cancels in this difference. The actual forced mean system has identical operator and shell rows for both preparations. Uniqueness therefore gives

$$\mathbf{U}_\lambda = \mathbf{U}_{\mathrm{base}} + \lambda\mathbf{W}, \qquad \mathbf{W}_{xx} - \mathbf{W}_{rr} = A_0\mathbf{W}_x + B_0\mathbf{W}_r + D_0\mathbf{W}, \quad L|_{r=0} = 0, \quad P_N\mathbf{W}_r = M_0 P_N \mathbf{W}. \tag{17}$$

Here P_N selects the other four slots. All metric–scalar, connection, physical Robin, and normal-gauge couplings are included.

The proper scalar mean illustrates why absolute and difference responses must be distinguished. In the background orthonormal shell frame $e_0 = \partial_\tau$, $e_a = A^{-1}E_a$, with E_a unit-round, write the first shell metric components as $m_{00} = -2XY$, $m_{0a} = \forall D_a Y$, $m_{ab} = 2(ZY\kappa_{ab} + dD_aD_bY)$ and the first scalar as ψY. With $C_\tau = X$, $C(0) = 0$, and $E = Z - HC$, its second proper mean is

$$M_2 = w_0 - C\psi_\tau/3 + (E - 8d/3)\psi. \tag{18}$$

The last two terms are clock and volume-measure corrections. Every first quantity here is shared; these terms cancel only on subtraction.

The shell Gauss contraction [6] gives $R(\gamma) = (D\phi)^2 + F(\phi)$, where $F = 2\mathcal{V} - (\sigma')^2/4 + \sigma^2/3$. In

the anchored foliation its volume-weighted second identity is

$$\begin{aligned}(v_2)_{\tau\tau}+4H(v_2)_\tau-2v_2/A^2&=\tfrac12F_b'M_2+Q_1,\qquad v_2(0)=(v_2)_\tau(0)=0,\\ Q_1&=F_b''\psi^2/12-\psi_\tau^2/6+4\psi^2/(3A^2)-5E^2/(3A^2)\\ &\quad+E_\tau^2-16E_\tau D_\tau/3+8D_\tau^2/3,\qquad D_\tau=d_\tau-\mathsf{V}/A-C/A^2.\end{aligned}\tag{19}$$

Thus every first-only quadratic term Q_1 also cancels in the difference. The absolute base response retains the full forcing, the corrections in (18), and every term of Q_1. No source is removed from that response to obtain the positive witness.

Put $m(x)=W_w(x,0)$. Equations (18)–(19) give the exact affine identities

$$M_{2,\lambda}=M_{2,\mathrm{base}}+\lambda m,\qquad v_{2,\lambda}=v_{2,\mathrm{base}}+\lambda\nu,\qquad \Theta_{2,\lambda}=\Theta_{2,\mathrm{base}}+\lambda\theta,\quad \theta=\nu_\tau.\tag{20}$$

There is no extra factor $1/2$ in these second-derivative slopes. The exact target functionals are

$$\begin{aligned}\mathcal{L}_M[h]&=m(T),\\ \mathcal{L}_v[h]&=(\rho_b\ell)^2(F_b'/2)\int_1^T k(T,x)m(x)\,\mathrm{d}x,\\ \mathcal{L}_\Theta[h]&=\rho_b\ell(F_b'/2)\int_1^T k_T(T,x)m(x)\,\mathrm{d}x,\\ k(T,x)&=\frac{\sinh(2\ell(T-x))}{2\ell}\frac{\cosh^2(\ell x)}{\cosh^2(\ell T)}.\end{aligned}\tag{21}$$

$F_b'<0$ on the selected background. The homogeneous solutions $\tanh(\ell x)$ and $1+\tanh^2(\ell x)$ give this kernel. For $v=\ell T$, $b=\ell x$, $k_T=\cosh^2 b\cosh(v-2b)/\cosh^3 v$. Therefore

$$0\le k\le T-x,\quad 0<k_T\le1,\qquad 1-\eta\le k/(T-x)\le1,\quad 1-\eta\le k_T\le1,\quad \eta=10^{-11}.\tag{22}$$

The ratio extends continuously at $x=T$. Lower bounds $(1+v^2)^{-2}$ and $(1+v^2)^{-3}$ suffice since $3v^2<\eta$.

4 A full coupled response certificate

Fix the single direction

$$\chi=\ell\rho_b^3Q(r),\qquad h=-(\rho_b/\rho)^3Q_r(r).\tag{23}$$

It satisfies (12) identically, without numerical moment cancellation. Define

$$R=(\rho_b/\rho)^3,\quad b=\ell\rho a,\quad c=-\frac{3\rho_b^3}{2\ell\rho^4p}>0,\quad d_0=\ell\rho p_y/p.\tag{24}$$

Primes in the following formulas mean r derivatives:

$$\begin{aligned}&h=-RQ',\quad h_r=-R(Q''+3bQ'),\quad V_L=-R(Q''+4bQ'),\\ &s=c(Q''+4bQ'+3\ell^2Q),\\ &s_r=c\{Q'''+4bQ''+3\ell^2Q'+(4b+d_0)(Q''+4bQ'+3\ell^2Q)-4\ell^2\rho^2(a^2+a_y)Q'\}.\end{aligned}\tag{25}$$

These use $\rho_r=-\ell a\rho^2$, $R_r=3bR$, $c_r=c(4b+d_0)$, and $b_r=-\ell^2\rho^2(a^2+a_y)$. The initial Gram norms for $\mathsf{H}=\mathrm{diag}(1,2,1,3,1)$ are $|\mathbf{W}_0|_{\mathsf{H}}=\sqrt{4h^2+s^2}$, $|\mathbf{W}_{0,r}|_{\mathsf{H}}=\sqrt{4h_r^2+s_r^2}$, and $|\mathbf{W}_{0,x}|_{\mathsf{H}}=\sqrt{2}|V_L|$.

On $D_T = \{0 \le x \le T,\ 0 \le r \le T - x\}$ use

$$X(\mathbf{U}) = \max\{\sup |\mathbf{U}|_{\mathsf{H}}/3, \sup |\mathbf{U}_x|_{\mathsf{H}}, \sup |\mathbf{U}_r|_{\mathsf{H}}\}. \tag{26}$$

The retained coefficient caps are $\|A_0\|, \|B_0\| \le 10^{-4}$, $\|D_0\| \le 10^{-6}$, $\|M_0\| \le 1/2000$. With $\Delta = T - 1 = 1/4$, the reflected bulk Duhamel operator and Robin lift have bounds $T \sup |F|_{\mathsf{H}}$ and $\sup |G|_{\mathsf{H}}$. Their combined contraction is

$$q = T(2 \cdot 10^{-4} + \Delta 10^{-6}) + \Delta/2000 = 1201/3200000 < 1. \tag{27}$$

Let $\mathbf{B}$ be the free reflected wave of (16) and let $D_0^{\mathrm{init}}, D_r^{\mathrm{init}}, D_v^{\mathrm{init}}$ bound its initial value, radial derivative and velocity Gram norms on $0 \le r \le T$. Then

$$N = \max\{(D_0^{\mathrm{init}} + TD_v^{\mathrm{init}})/3, D_r^{\mathrm{init}} + D_v^{\mathrm{init}}\} \ge X(\mathbf{B}), \qquad E_h = \frac{qN}{1-q} \ge X(\mathbf{W} - \mathbf{B}). \tag{28}$$

This is the full resolvent, not a truncated iteration or an uncoupled scalar approximation. The common actual forcing cancels exactly in (17); no source-history budget is needed for this difference.

Both waves vanish in $x + r \le 1$. At the shell, even scalar reflection and zero scalar initial velocity give $B_w(x, 0) = s(x)$. Hence

$$|m(x) - s(x)| \le (x-1)_+ E_h, \qquad |m(T) - s(T)| \le \Delta E_h. \tag{29}$$

Define free moments $J_v = \int_1^T (T - x)s(x)\,\mathrm{d}x$ and $J_\Theta = \int_1^T s(x)\,\mathrm{d}x$, with absolute versions A_v, A_Θ. The kernel-weighted histories differ from these by at most

$$\frac{\Delta^3 E_h}{6} + \eta A_v, \qquad \frac{\Delta^2 E_h}{2} + \eta A_\Theta, \tag{30}$$

respectively. This proves continuous-history bounds without assuming a pointwise sign of the evolved scalar arbitrarily close to first contact.

The registered interval calculation uses the fixed $N = 1024$ partition restricted to this causal initial segment: one grouped flat tail $[1, 65/64]$ and 240 interior cells $[k/1024, (k+1)/1024]$, $k = 1040, \ldots, 1279$. It makes 241 whole-cell bump evaluations and one distinct bump target evaluation at $r = 5/4$, 242 in total, with zero new background or evolved-first-field oracle calls. The segment $[0, 1]$ is exactly zero. The flat tail uses the monotone bounds for $s_0^{2j} e^{4-s_0}$ with $s_0 \ge 4096/63$, $j \le 3$, including the flat endpoint. Positive exact cell weights integrate the whole-cell scalar boxes. Fresh profile bounds are used; old preparation profiles and old total-response norms are not substituted for (25).

Outward upper displays are

$$D_0^{\mathrm{init}} \le 3.0797\,10^9, \quad D_r^{\mathrm{init}} \le 4.5253\,10^{10}, \quad D_v^{\mathrm{init}} \le 46.892, \quad N \le 4.5253\,10^{10}, \quad E_h \le 1.6991\,10^7. \tag{31}$$

The exact rational arithmetic in (21)–(30) gives

Full coupled slope	Outward enclosure
$\mathcal{L}_M[h]$	$[1.5393, 1.7068]\,10^9$
$\mathcal{L}_v[h]$	$[-2.1094, -1.5795]\,10^{-4}$
$\mathcal{L}_\Theta[h]$	$[-22.9410, -17.4038]$

The nonzero observable slopes are thus certified, rather than inferred from non-gauge initial data. Normal and optimized runs and an independent reconstruction agree in their rational scientific

outputs [9]. They share analytic premises and are not independent observations of the nonlinear PDE. Appendix A supplies a broad sign proof without the new direction quadrature or bump oracles.

For base lower endpoints $b_v^-, b_\Theta^- < 0$ and slope upper endpoints $l_v^+, l_\Theta^+ < 0$, choose the least nonnegative integer n such that

$$L = 10^n > \max\{(-b_v^-)/(-l_v^+), (-b_\Theta^-)/(-l_\Theta^+)\}, \qquad \lambda = -L. \tag{32}$$

This algebraic rule was fixed before the direction evaluation. It alters no direction, normalization, partition, target, or sign gate. The exact threshold lies below $5.184\,10^{20}$ and above 10^{20}, so it selects $L = 10^{21}$. Substitution in the exact affine relations (20) proves the positive intervals in Theorem 1. The coarse certificate independently selects the same power and proves $v_{2,-10^{21}} > 7\,10^{16}$ and $\Theta_{2,-10^{21}} > 5\,10^{21}$.

5 Meaning, limits, and relation to prior work

At this clock $V_0 > 0$ and $\Theta_0 = 3\rho_b^{-1}\tanh(\ell T) > 0$. All first integrated coefficients vanish by the shared quadrupole mean. For each of the two fixed preparations, C^2 parameter dependence gives

$$\frac{V_\varepsilon}{V_0} = 1 + \tfrac{1}{2}\varepsilon^2 v_2 + o(\varepsilon^2), \qquad \Theta_\varepsilon = \Theta_0 + \tfrac{1}{2}\varepsilon^2\Theta_2 + o(\varepsilon^2). \tag{33}$$

For sufficiently small nonzero ε of either sign the base preparation therefore has smaller positive volume and slower positive expansion than the background. The second preparation has larger volume and faster positive expansion. Both families remain expanding. A common qualitative interval for these two fixed families is obtained by taking the smaller of their unknown intervals. Opposite second coefficients do not establish contraction, an effective physical amplitude, or a finite-time sign history. Large second coefficients in this normalization are not finite-amplitude field values, and no SI calibration is supplied.

Each nonzero slope permits arbitrary individual real second coefficients as fixed finite λ varies. This one line reverses both signs because its slopes have the same sign. It does not prove rank two or arbitrary independent pairs (v_2, Θ_2). Directions outside the target causal region can be invisible there. The result also supplies no full angular waveform, physical packet-generation process, hot matter, radiation epoch, or evidence concerning a cosmological origin.

The general homogeneous freedom and affine second-response structure are established machinery, not a new principle. Minamitsuji, Himemoto and Sasaki explicitly retain initial-condition information in a second-order scalar-induced Weyl term [1]; their scalar does not couple to the tension and their late-time regime differs. Deffayet's linear master initial-boundary analysis [2] explains why brane rows do not replace bulk initial data. Hiramatsu and Koyama explicitly caution that their curvature spectrum uses a special bulk preparation [3]. These works do not certify the compact direction or the proper-volume sign here. A recent AdS_3/BCFT model illustrates classical trajectory-data freedom [7]; another proposes Euclidean no-boundary regularity as a preparation prescription [8]. Both have a brane-local scalar and pure bulk gravity, so their prescription cannot be imported into this bulk-scalar model without a separate derivation.

The contribution is a conditional, model-specific quantitative counterexample to simultaneous sign universality under the stated first data. The focused primary-source comparison is not exhaustive and does not establish originality or priority. The original result for its expressly selected preparation remains valid. A predictive blast model requires an additional physical rule selecting the second preparation; this calculation neither supplies that rule nor disproves every blast model. Other models require their own equations, initial conditions, and observables.

A Elementary sign certificate without direction quadrature

Lemma 1 (Coarse sufficient bounds)**.** *For direction (23), assume the retained coefficient bounds and the following uniform exact caps on the causal initial segment:*

$$\begin{aligned} 129 \le \rho, \rho_b \le 131, \quad 3/25000 \le -p \le 1/7000, \quad 111/1000 \le a \le 14/125, \\ |p_y| \le 1/4000, \quad |a_y| \le 1/10000, \qquad 1/1200 \le -F_b' \le 1/1000. \end{aligned} \tag{34}$$

Then the full coupled slopes obey

$$\mathcal{L}_M[h] \in [1.269\,10^9, 2.198\,10^{10}], \quad \mathcal{L}_v[h] \in [-0.005883, -0.0001282], \quad \mathcal{L}_\Theta[h] \in [-359.5, -14.21]. \tag{35}$$

The same fixed $\lambda = -10^{21}$ makes both second coefficients positive.

Proof. For $t = r - 1$, $D = t(1-t)$, on $1 < r \le 5/4$ direct differentiation gives

$$Q' = \frac{Q(1-2t)}{D^2}, \quad Q'' = \frac{Q(1-6D+6D^2)}{D^4}, \quad Q''' = \frac{Q(1-2t)(1-10D+24D^2-12D^3)}{D^6}. \tag{36}$$

Here $0 < D \le 3/16$. The second numerator decreases from 1 to $11/128 > 0$; the cubic numerator decreases from 1 to $-113/1024$. Its absolute value is at most 1. Set $w = 1/D \ge 16/3$. For powers 2 and 4, $e^{-w}w^j$ decreases on this interval; for power 6 its maximum occurs at $w = 6$. Rational Taylor bounds $1/4 < e^{-4/3} < 4/15$ and $e^{-2}6^6 < 6400$ give, continuously through the flat endpoint,

$$0 \le Q \le 1/3, \quad |Q'| \le 8, \quad |Q''| \le 216, \quad |Q'''| \le 6400. \tag{37}$$

No unresolved sampled tail enters these bounds. Moreover Q, Q', Q'', and hence s, are positive for $1 < r \le T$. With $c_{\min} = 22540234500000000/294499921$, the caps imply $c \ge c_{\min}$. The exact endpoint values are $Q(T) = e^{-4/3}$, $Q'(T) = (128/9)e^{-4/3}$, and $Q''(T) = (5632/81)e^{-4/3}$. Positivity and integration by parts yield

$$J_v > c_{\min}/4, \qquad J_\Theta > 32c_{\min}/9, \qquad s(T) > 1408c_{\min}/81. \tag{38}$$

These avoid cancellation in the free moments.

For reproducible upper bounds put

$$\begin{aligned} R_+ = (131/129)^3, \quad b_+ = \ell\,131(14/125), \quad d_+ = \ell\,131(1/4000)/(3/25000), \\ c_+ = 28101137500000000/276922881, \\ e_+ = \ell^2\,131^2[(14/125)^2 + 1/10000], \qquad S_+ = c_+[216 + 32b_+ + \ell^2], \\ S_{r,+} = c_+\{6400 + 864b_+ + 24\ell^2 \\ + (4b_+ + d_+)[216 + 32b_+ + \ell^2] + 32e_+\}. \end{aligned}$$

Equations (25), with the triangle inequalities $\sqrt{4h^2+s^2} \le 2|h| + |s|$ and $\sqrt{2}|V_L| \le 2|V_L|$, permit

$$D_0^{\text{init}} \le 16R_+ + S_+, \quad D_r^{\text{init}} \le 2R_+(216 + 24b_+) + S_{r,+}, \quad D_v^{\text{init}} \le 2R_+(216 + 32b_+).$$

Substitution in the exact (28) gives $E_h \le 243841044.434$. The strict positive gates for the coupled histories are

$$m(T) > 1408c_{\min}/81 - \Delta E_h > 0, \quad \int_1^T km > (1-\eta)c_{\min}/4 - \Delta^3 E_h/6 > 0, \tag{39}$$

and $\int_1^T k_T m > (1-\eta)32c_{\min}/9 - \Delta^2 E_h/2 > 0$. The latter uses the exact upper bound $k_T \le 1$ for the feedback error. The matching upper bounds use (37) and the same errors. All are rational operations apart from the stated rational exponential bounds. Inserting (34) into (21) yields (35). The exact coarse threshold in (32) lies between 10^{20} and $6.345\,10^{20}$, selecting 10^{21} and giving $v_{2,-10^{21}} > 7\,10^{16}$ and $\Theta_{2,-10^{21}} > 5\,10^{21}$. □

The standalone coarse arithmetic reads only the two pinned retained background/base-response JSON inputs. It calls no bump, background, or evolved-first-field oracle and performs no direction quadrature. Its agreement with the tighter calculation is an independent bounded method under the same analytic and background premises, not a new PDE observation. Source pins, the exact coarse report, the frozen whole-cell report, decimal enclosure checks, and build receipt accompany this manuscript [9].

References

[1] M. Minamitsuji, Y. Himemoto, and M. Sasaki, *Geometry and cosmological perturbations in the bulk inflaton model*, Phys. Rev. D **68**, 024016 (2003), arXiv:gr-qc/0303108.

[2] C. Deffayet, *A note on the wellposedness of scalar brane world cosmological perturbations*, Phys. Rev. D **71**, 023520 (2005), arXiv:hep-th/0409302.

[3] T. Hiramatsu and K. Koyama, *Numerical study of curvature perturbations in a brane-world inflation at high-energies*, JCAP **12** (2006) 009, arXiv:hep-th/0607068.

[4] M. Bruni, S. Matarrese, S. Mollerach, and S. Sonego, *Perturbations of spacetime: gauge transformations and gauge invariance at second order and beyond*, Class. Quantum Grav. **14** (1997), arXiv:gr-qc/9609040.

[5] M. Mars, *First and Second Order Perturbations of Hypersurfaces*, Class. Quantum Grav. **22** (2005), arXiv:gr-qc/0507005.

[6] T. Shiromizu, K. Maeda, and M. Sasaki, *The Einstein Equations on the 3-Brane World*, Phys. Rev. D **62**, 024012 (2000), arXiv:gr-qc/9910076.

[7] K. Fujiki, H. Kanda, M. Kohara, and T. Takayanagi, *Brane Cosmology from AdS/BCFT* (2025), arXiv:2501.05036, Sec. 5.5.

[8] K. Fujiki, M. Kohara, J. Kristiano, Y. Suzuki, and T. Takayanagi, *Effective Dynamics of Inflationary End-of-the-World Branes in AdS_3* (2026), arXiv:2609.11643.

[9] R. Maldonado, *HDBLAST conditional technical records and reproducibility materials* (2026), repository. Retained records include the localized initial-data and common-gauge development, the 1 November prepared-shell volume capsule, and the 4 November preparation-difference capsule accompanying this working paper. The latter contains the frozen experiment, exact rational outputs, independent reconstructions, elementary sign certificate, and focused primary-source literature comparison.